Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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September 07, 2026

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Abstract

Background. Treatment response in aggregated N-of-1 clinical trials reflects three causally distinct mechanisms: pharmacological action of the active compound (BR), the patient's belief that the treatment will work (PB), and natural-history drift in the underlying condition (TV). Standard analysis models combine these into a single 'treatment effect' plus residual error. Whether this single-indicator combination biases inference is not unconditional. That is, it depends on the estimand and on whether candidate biomarkers correlate with the non-pharmacological components, a dependence we make precise in this paper.

Methods. We formalize a three-component decomposition of the within-participant outcome trajectory, each component a modified Gompertz function with participant-specific random effects, and we characterize the trial-design contrasts (open-label, blinded discontinuation, crossover) that supply identifiability. We derive the omitted-variable-bias identity for the one-component estimator, which expresses both the one-component treatment effect and the one-component biomarker slope as sums of component-specific terms. We then evaluate the analysis strategies in a pre-registered Monte Carlo study (Study A) calibrated to the prazosin-PTSD reference parameters, comparing a one-component design-contrast analysis with a phase-augmented analysis at 1,000 replicates per alternative cell (5,000 per null cell) across $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 200, 300\}$.

Results. The identity shows that the one-component biomarker-treatment slope is displaced from the pharmacological slope by a term proportional to the biomarker's covariance with PB and TV, not by the magnitude of those components. Study A confirms this. With a biomarker coupled to BR only, the one-component estimate of the biomarker-treatment interaction is essentially unbiased across the whole grid (mean absolute bias 0.016 at $N = 35$ and 0.006 at $N = 300$, within one Monte Carlo standard error (MCSE) of zero), with near-nominal coverage and Type I error. The phase-augmented analysis provides no bias reduction and is markedly less powerful (mean power 0.35 versus 0.59 at $N = 35$; 0.62 versus 0.90 at $N = 70$), so it is dominated under this uncontaminated biomarker. A contaminated-biomarker pilot then exhibits the failure mode directly: coupling the biomarker to PB drives the one-component bias up with the coupling strength, reaching +0.51 at a biomarker-PB correlation of 0.45 ($N = 300$, some 30 MCSEs from zero). The operative quantity is the biomarker's correlation with the individual-level PB *variance*, not the population-mean placebo response, which does not enter the slope; the phase-augmented analysis does not remove this bias either. We conclude that the inferential value of decomposition is conditional, not universal.

Conclusions. The trial design must supply the drug-on/off and belief contrasts. Given those, component-level decomposition is useful when the estimand is the attribution of response to pharmacology versus belief versus natural history, or when a candidate biomarker may correlate with PB or TV. For a biomarker-treatment interaction with a biomarker orthogonal to PB and TV, a single-indicator analysis that exploits the design contrast is sufficient, and a parametric component model is counterproductive. A pragmatic specification for the cases that warrant it is `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase` with random intercept and AR(1) residual correlation. A contaminated-biomarker study demonstrates the failure mode: the one-component slope is biased in proportion to the biomarker's correlation with PB. Study B shows that the bias is removable, but only under a design that decouples drug exposure from belief. On a balanced-placebo design the belief-covariate decomposition recovers the unbiased pharmacological slope (bias within one MCSE of zero across the feasible contamination range at N=300), while the one-component estimator does not; on the standard hybrid design no decomposition recovers it. Recovery is therefore conditional on the trial design, not on the analysis alone.

1 Introduction

1.1 A claim about how N-of-1 trials should be analyzed

We begin with an opinionated methodological claim. Treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components, and it should routinely be analyzed as such, rather than estimated as a single one-component quantity. The three components are pharmacological action (BR, the biological response), placebo-belief response (PB, the patient's belief that they are receiving treatment), and time-variant natural history (TV, the trajectory that would have occurred without intervention). Each is causally distinct. Each is independently of clinical and regulatory interest, and each is identifiable from a properly designed N-of-1 trial. We shall argue that the standard practice of fitting a single mixed-effects model with one treatment indicator, prevalent across the published N-of-1 literature, conflates these three components. That practice biases the interpretation of treatment effect, and it renders predictive-biomarker validation unanswerable as a scientific question.

We make no claim of novelty for the decomposition itself. The structural decomposition of a longitudinal treatment response into separable disease-progression, placebo, and drug-action components is a well-established tradition in pharmacometrics[1–3]. The separation of drug from placebo response in longitudinal trajectories has been pursued both through growth-mixture models[4] and through experimental decomposition of the placebo response itself[5]. Each ingredient we use, namely mixed-effects models with trial-phase indicators, placebo-arm contrasts, regression-to-mean adjustment, and growth-curve parameterization, has likewise appeared across decades of clinical-trial methodology. Our contribution is therefore one of consolidation and extension rather than invention. We make the three-component structure explicit for the aggregated N-of-1 setting, we add a separately modeled placebo-belief channel identified by design contrasts rather than by functional form alone, and we argue that the decomposition should become routine wherever a longitudinal treatment response is analyzed. We document why the framing has not been routine in N-of-1 practice, citing parsimony, identifiability concerns, sample-size constraints, and methodological inertia, and we propose an analysis that is tractable at trial-relevant sample sizes through a phase-augmented linear-mixed-effects approximation to the full decomposition.

1.2 Why this matters: three failure modes of the one-component analysis

We begin with the failure modes of the standard approach. A one-component analysis, in which the treatment effect is taken to be what we observe minus baseline, has three identifiable failure modes that arise when the population displays meaningful heterogeneity in any of the unmodeled components. None of these is hypothetical; each has a documented presence in the chronic-condition trial literature. Two points of precision, established formally in section 6.1 and borne out by the simulation of section 7.3, should be carried through what follows. The first two failure modes attach to the baseline-anchored one-component effect, and they are removed once the design supplies a drug-on/off contrast, which the one-component analysis of later sections in fact uses. The third failure mode, biomarker-validation failure, is different in kind. It requires not merely that PB or TV be large but that the candidate biomarker correlate with them, and where that coupling is absent the one-component analysis is not biased at all.

The first is the placebo-attribution failure. When the population contains both high and low placebo responders, the one-component treatment effect overestimates the pharmacological component. The bias scales with the magnitude and variance of PB in the study population, and it is invariant under sample-size increases: a larger one-component trial converges on the same biased estimate. The most consequential example in modern clinical pharmacology is antidepressant efficacy.[6] analyzed FDA-submitted data on four widely-prescribed antidepressants and showed that, except at the most severe baseline-severity stratum, mean drug-placebo differences fell below clinical-significance thresholds, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation.[7] reproduced this pattern in late-life depression and explicitly invoked regression to the mean, and[8] generalized the finding to anxiety. Patient populations differ systematically in placebo responsiveness, and the difference has measurable biological correlates: COMT val158met genotype predicts placebo response in irritable bowel syndrome[9,10], indicating that 'placebo' is not a homogeneous nuisance but a structured biological phenomenon that interacts with the patient's genome and treatment history. A trial whose population happens to be placebo-responsive will report a larger 'drug effect' than one whose population is not, even when the underlying pharmacology is identical.

The second failure mode is regression to the mean. When patients enrol at moments of clinical engagement that coincide with peak symptom severity, natural-history regression toward typical severity contributes several points of apparent improvement in the early weeks of any trial, without any pharmacological contribution at all. The systematic reviews of[11],[12], and[13], the last covering 202 trials across 60 clinical conditions, quantified this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The one-component analysis absorbs this regression into the treatment-effect estimate, and the analyst has no formal mechanism to recover the true pharmacological effect.

The third failure mode is the biomarker-validation failure. The precision-medicine question that motivates many recent N-of-1 trials is whether a baseline biomarker predicts treatment response, formalized in the PATH statement[14,15]. A biomarker that correlates with placebo responsiveness or with a favorable natural-history trajectory will appear in a one-component analysis as a predictor of the one-component 'treatment effect', indistinguishable from a true pharmacological predictor. The biomarker-stratified prescribing decision that follows from such an analysis is wrong, and the trial's regulatory utility is degraded. The PATH framework explicitly defers the mechanism of treatment-effect heterogeneity to domain knowledge[16,17]; the BR-PB-TV decomposition supplies the mechanistic substrate that PATH leaves blank. We argue that, whenever the candidate biomarker may itself correlate with PB or TV, predictive-biomarker validation cannot be conducted rigorously in a one-component analysis framework regardless of how large the trial is; the bias is fixed by the biomarker's coupling to the non-pharmacological components, not by sample size. Where the biomarker can be assumed orthogonal to PB and TV, this failure mode does not arise, and the one-component analysis recovers the pharmacological slope, as the simulation of section 7.3 confirms.

We stress at the outset the precise form this failure mode takes, because the paper's own simulation history clarifies it. Study A (section 7.3) varies the *magnitude* of PB and TV with the biomarker coupled to BR alone, which the section 6.1 identity predicts to be innocuous, and it finds no bias. The decisive experiment is the contaminated-biomarker pilot (section 7.4), which couples the biomarker to PB and exhibits the failure directly: the one-component bias grows with the biomarker-PB correlation. The operative quantity is the biomarker's correlation with the individual-level PB *variance* ($c_{bm,PB}\,\sigma_{PB}$), not the population-mean placebo response m_{PB}, which does not enter the slope. A trial population can have a large average placebo response and yet pose no validation hazard, while one with zero average but a biomarker-correlated spread of placebo responsiveness is at risk. The failure-mode claims of this section are therefore consequences of the identity, demonstrated in simulation (section 7.4), and the bias is recoverable: Study B (section 7.5) shows that a belief-covariate decomposition recovers the unbiased pharmacological slope under contamination, but only on a design that decouples drug exposure from belief, not on the standard hybrid design.

1.3 The current state of the N-of-1 literature

The N-of-1 trial in its modern randomized form originates with[18], who defined the within-patient multiple-crossover RCT, demonstrated it on a single airflow-limitation case, and established a clinical service to deliver it at scale. The design inherits its identifying logic from two older traditions: the crossover trial of classical biostatistics, in which within-subject period contrasts estimate treatment effects under explicit carryover assumptions, and the single-case experimental designs of behavioral science, in which phase reversals (withdrawal and reinstatement of treatment) identify effects within one participant. The blinded-discontinuation contrast central to this paper is a direct descendant of both. The methodological step from the individual-patient trial to population-level inference was taken by[19] and[20], who showed that a series of single-patient trials can be combined through hierarchical meta-analysis to estimate both population treatment effects and individual responses, the statistical foundation on which the aggregated design rests. Over the last fifteen years the design has accordingly been reframed from an individual-patient decision tool into an aggregated, meta-analytic framework supporting population-level inference: the programmatic statement of[21], the methodological reviews of[22], the historical-practice audit of[23], and the more recent syntheses of[24],[25],[26], and[27] trace that trajectory. Reporting standards are codified in the CONSORT extension[28] (the headline statement) and[29] (the explanation-and-elaboration companion); the reporting-quality audits of[30] and its successors document slow but progressive improvement in analytic transparency. Three observations about this literature motivate the present paper.

The first observation is that the dominant analytic models are simple. The reporting-quality audit of[30] documents that the published N-of-1 trials between 1985 and 2013 relied predominantly on paired t-tests, simple linear mixed-effects models with a binary treatment indicator, or hierarchical Bayesian regressions. On our own reading of that corpus and the methodological literature that followed it, no published N-of-1 trial has employed a structural decomposition of treatment response into pharmacological, expectancy, and natural-history components. The most recent methodological advances retain this character:[31] develops Bayesian distributed-lag models with autocorrelated errors as a current carryover-aware analysis, but the model still treats the response as a single quantity to be estimated.[32] and[33] compare aggregated N-of-1 trials with parallel and crossover RCT alternatives by simulation and document carryover-induced Type I error inflation as a key failure mode, but again at the level of the one-component treatment effect. None of these papers decompose the response.

The second observation is that the placebo-component separation problem is widely known but largely unaddressed in N-of-1 specifically. The blinded discontinuation phase used in the Hendrickson hybrid design[34] is one of the few protocol-level mechanisms in routine N-of-1 use that supplies identifiability of PB versus BR. Other relevant approaches, including placebo run-in designs, open-label-placebo trials, and mediator-analysis applications, have been developed in parallel but rarely imported into N-of-1 methodology.

The third observation is that the closest precedent in the literature is Kaptchuk's three-component decomposition of the placebo response itself. In a randomized trial of acupuncture-style placebo for irritable bowel syndrome,[5] decomposed the placebo response into observation, ritual, and patient-practitioner relationship components, demonstrating empirically that the placebo response is itself a sum of identifiable parts rather than a single 'expectancy' lump. The follow-up open-label placebo trials of[35],[36], and[37] in adolescent functional pain establish that even disclosed-placebo intervention can produce clinically meaningful PB-only response. The framework we propose is structurally analogous to Kaptchuk's decomposition but applied at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone. To the best of our knowledge, no published paper proposes the corresponding decomposition for the aggregated N-of-1 setting, even though the methodological ingredients, including mixed-effects modeling, blinded-phase contrasts, and trajectory parameterization, are all in routine use.

The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the failure modes of the one-component analysis are large enough in chronic-condition pharmacology to make the decomposition a methodological priority rather than an academic refinement.

1.4 The neurobiological and epidemiological reality of the components

The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures, and we shall briefly recall each in turn before turning to the formal model.

The biological response component represents the pharmacological action of the active compound: the chemical binding, downstream signaling, and physiological response that a regulatory authority cares about when it grants approval. The placebo-belief response reflects neurobiologically real mechanisms documented across pain, depression, Parkinson's, and PTSD research[38–41], mediated by belief-driven release of endogenous opioids and dopamine, anticipatory regulation of the hypothalamic-pituitary-adrenal axis, and top-down attentional shifts in symptom perception. Direct neuroimaging evidence is instructive here:[42] developed an fMRI-based 'neurologic pain signature' (NPS) that tracks nociceptive input and is reduced by the opioid agonist remifentanil, and the individual-participant meta-analysis of[43] found that placebo treatments produce moderate reductions in reported pain but only negligible reductions in the NPS, providing a neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. The placebo response is heterogeneous across patients with biological correlates[9,10] and is modulated by trial phase: when belief varies, as it does between open-label and blinded-discontinuation conditions, the placebo response varies, and the modulation is exactly the design contrast that makes the decomposition identifiable.

The time-variant natural-history component reflects whatever trajectory the participant would have followed without treatment. For acute conditions this trends positive, that is, recovery on an intrinsic timescale; for chronic stable conditions it is near zero on the population mean but with substantial participant-level variance; for chronic progressive conditions it trends negative; and for selection-driven enrollment processes it is positive in the early weeks of the trial regardless of underlying disease class, because of regression to the mean[11,13]. The PATH statement[14,15] highlights that this heterogeneity is itself the central object of precision-medicine inference: HTE must be modeled, not absorbed into noise, if the precision-medicine question is to be answered.

1.5 What this paper contributes

We make four contributions, each addressed in a section below. We begin in section 2 with the formal model and notation. Section 3 develops each component in turn, with biological motivation, parametric form, and identifiability conditions. Section 4 maps the design-contrast structure of the standard hybrid open-label-plus-blinded-discontinuation design onto the identifiability requirements of the decomposition. Section 5 develops the analysis-side methodology, including the linear-mixed-effects approximations and the phase-by-treatment indicator extension that retains BR-PB sensitivity at low parametric cost. Section 6 presents the worked examples, and section 7 presents the simulation study. Section 8 discusses applications to predictive-biomarker validation. The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` provides an extended worked-example treatment for readers approaching the material for the first time.

2 Notation and the formal model

In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at `docs/24-component-decomposition-pedagogy.md`.

For participant i at trial timepoint t, we may write:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[\, BR_{it} + PB_{it} + TV_{it} \,\right] \;+\; \varepsilon_{it},$$

Each component is a participant-specific function of its own time-on-state variable. BR_{it} depends on time-on-drug, the cumulative exposure to active medication since drug initiation. PB_{it} depends on time-on-belief, modulated by a phase-specific expectancy factor $\eta(\text{phase})$ that captures the patient's confidence that they are receiving active treatment. TV_{it} depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

Each component follows a modified Gompertz curve. In symbols:

$$C(t) \;=\; m_C\,\exp(-d_C\exp(-r_C t)) \;-\; m_C\,\exp(-d_C),$$

ε_{it} is a within-participant residual term with AR(1) serial-correlation structure $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2\rho^{|h|}$. We note that the three components are permitted to be jointly correlated through within-time and across-time cross-component covariances c_{cf1t} and c_{cfct} respectively, so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

The principal psychometric and pharmacological terms used below are summarized in Table 1.

Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

Term	Meaning
BR	Biological response: pharmacological reduction in symptom score attributable to active drug.
PB	Placebo-belief response: reduction attributable to the patient's belief that they are receiving treat
TV	Time-variant component: reduction (or increase) attributable to the natural-history trajectory of
η(phase)	Expectancy multiplier for PB: 1 in open-label on-drug, ≈ 0.5 in blinded discontinuation, between
BL_i	Participant i's baseline (pre-treatment) symptom score.
D_{it}	Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$.

3 The three components

In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterization, and the design contrasts that identify it. The treatment is organized so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

3.1 Biological response (BR)

BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR.

The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life, and brain concentration follows plasma with a short delay. The downstream pharmacodynamic response, that is, the reduction in symptom-relevant neural activity in the case of prazosin's effect on noradrenergic tone during REM sleep, typically requires several days to weeks of sustained exposure to integrate fully[44]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

The modified Gompertz form $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR} t)) - m_{BR} \exp(-d_{BR})$ captures this profile compactly. The asymptote $m_{BR}(1-e^{-d_{BR}})$ is the biological ceiling, the largest reduction the drug can produce at infinite exposure. The scale parameter m_{BR} equals that ceiling up to the factor $(1-e^{-d_{BR}})$, which for the calibrated $d_{BR} \approx 5$ exceeds 0.99. The onset rate r_{BR} controls how rapidly the participant approaches the ceiling: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} shapes the climb: high d_{BR} gives an early-onset curve that rises sharply at first, and low d_{BR} a more gradual climb. For PTSD prazosin, calibrated values from the[34] reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (moderately steep early climb).

Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1-\exp(-rt))$ is acceptable for fast-onset drugs, but it lacks the slow-start phase that characterizes most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high d limit) and linear-in-t growth (low d limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting.

The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modeling. The sigmoid $E_{\max}$ (Hill) model[45] describes the dependence of effect on concentration, that is, a dose-response curve, whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as `07-gompertz-evaluation`. The headline result of that evaluation is reassuring for the present framework: in a 1{,}000-replicate-per-cell simulation that generated data under the logistic, hyperbolic-tangent, and piecewise-linear breakpoint families and analyzed them under the standard linear-mixed model, the trajectory family was essentially irrelevant to the power of the biomarker-treatment interaction test (the family-to-family spread fell within two MCSEs of sampling noise) and to Type I error (0.025 to 0.036 across null cells, with no family-specific inflation). The conclusions drawn here are therefore not artifacts of the Gompertz form, though the regimes the family cannot represent, namely cyclical or biphasic trajectories, remain genuine exclusions.

The BR component is identified by within-participant on-drug versus off-drug contrasts, formalized in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR decays toward zero as the drug clears while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of BR that does not require the analyst to model PB or TV explicitly. The decay is not instantaneous: with the carryover half-life $t_{1/2} = \ln 2/\lambda$ of the analysis-side decay term, residual BR persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to $t_{1/2}$, and the drug state is best carried by the continuous-decay indicator D_{it} of Table 1 rather than a hard on/off indicator.

3.2 Placebo-belief response (PB)

PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

It is worth being precise about what PB is and is not. PB is not 'fake' improvement; it is a phenomenon clinicians knowingly exploit, and a national survey of US internists and rheumatologists found roughly half report regular use of placebo treatments[46]. As the introduction sets out, the placebo response is mediated by well-characterized neurobiological pathways[38–41,47]: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the[42] neurologic pain signature[43] and shown to have genetic correlates[9,10]. What makes these mechanisms 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

For trial design, the relevant point is that PB is separately identifiable from BR if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

PB also follows a Gompertz curve, scaled by the expectancy factor. In symbols:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB} t)) \;+\; (\text{offset to start at zero}).$$

Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in blinded discontinuation, and η between 0 and 0.5 in open-label placebo (the residual expectation a patient retains when told the previous active phase was beneficial). The quantity η operationalizes response expectancy in the sense of[48], the patient's anticipation of benefit that the expectancy literature treats as a determinant of experienced symptom change. The choice of $\eta = 0.5$ for the blinded phase is a modeling assumption rather than a calibrated quantity: to our knowledge the residual expectancy a patient carries when uncertain about allocation has not been measured directly as a fraction of the open-label level, and we adopt 0.5 as a deliberately central placeholder. The available indirect evidence is nonetheless that the residual response is substantial. Open-label placebo trials[35–37] demonstrate that a clinically meaningful response persists even when patients are explicitly told they are receiving an inert intervention, which bounds the residual belief-driven response away from zero. Because the value of η is not empirically pinned, sensitivity analyses varying it across its plausible range are mandatory before any conclusion is allowed to depend on this single number.

The decomposition framework as proposed here has a structural analogue in the placebo-research literature.[5] decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomized trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone.

A subtle property of PB is that its variance also scales with η, not just its mean. When η is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-expectancy patients show large PB, and low-expectancy patients show essentially none. When η is low (blinded), the placebo response is uniformly attenuated and the between-participant variance shrinks accordingly. The model therefore parameterizes $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure deflates standard errors in the open-label phase and inflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

In this section we shall broach a question that the formal model leaves open: whether the three components contribute additively to symptom reduction, or whether there is a meaningful interaction between BR and PB that the additive specification misses.

The decomposition as written in section 2 treats the three components as additive contributors to symptom reduction. In symbols:

$$Y_{it} \; = \; \mathrm{BL}_i \; - \; (BR_{it} + PB_{it} + TV_{it}) \; + \; \varepsilon_{it}.$$

The additivity is a modeling assumption, not a fact about biology, and it is worth stating that the assumption may be wrong in directions the framework can absorb and in directions it cannot. Additivity is also scale-dependent in the sense emphasized by[49]: components that combine non-additively on the raw symptom scale may combine additively on a transformed (for example logarithmic) scale, so an apparent $BR \times PB$ interaction can be partly an artifact of the chosen response metric rather than a biological fact. That belief and pharmacology interact rather than act in isolation is demonstrated directly by[50], who showed with neuroimaging that positive treatment expectation roughly doubled the analgesic benefit of the opioid remifentanil while negative expectation abolished it, so the two channels are not in general separable by simple addition. More specifically, the published literature on placebo-drug interactions has accumulated evidence that the combined effect of receiving an active drug and believing one is receiving an active drug is, in some clinical contexts, less than the sum of the two components measured separately[5,38]. The phenomenon is typically termed subadditivity: $\partial Y/\partial(BR \cdot PB) > 0$ when both components are present and their interaction term is included in a saturated mean-structure model.

Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, a shared neural substrate: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, ceiling effects on the symptom scale: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose $BR + PB$ already saturates the scale will register no further reduction even if BR or PB would, in isolation, predict more. Third, belief-dynamics adaptation: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing PB contribution; equivalently, unexpectedly large BR may attenuate PB via expectation recalibration.

Whichever mechanism dominates in a given clinical context, the subadditivity question is empirical and design-dependent. Within the BR-PB-TV framework, subadditivity manifests as a non-zero coefficient on a $BR \times PB$ interaction term in the conditional-mean specification. The additive form treats this interaction as zero by construction; a generalized form, in symbols:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}\right] \;+\; \varepsilon_{it},$$

For trial design, two consequences follow. First, a biomarker that predicts BR in an additive-decomposition analysis may overstate or understate predictive accuracy if the true mechanism is subadditive: under subadditivity, the biomarker-driven increase in BR is partly cancelled by the $\gamma \cdot BR \cdot PB$ interaction, and the marginal biomarker-treatment association is smaller than the biomarker-by- BR association. Second, trial designs that attenuate PB in some phases (blinded discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by-BR association than is the open-label estimate. The phase-augmented analysis recommended in section 5 below therefore has the side benefit of partly absorbing subadditivity, even when γ is not estimated explicitly.

A simulation study quantifying the magnitude of bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero γ is forthcoming as part of the simulation program described in section 7. The pre-registration is that bias scales linearly with γ at small $|\gamma|$, with the sign opposite to the true biomarker-by-BR effect under subadditivity and aligned with it under superadditivity.

A subtle but inferentially important point merits attention here: a non-zero $\hat{\gamma}$ recovered from an additive-extension analysis does not, on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modeling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in `analysis/report/03-latent-class-mixture-application/`), and class membership correlates with both BR strength and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the one-component analysis perspective this looks exactly like a γ-weighted $BR \times PB$ interaction even though no such direct interaction was specified in the class-conditional DGP. That is, the latent-class structure is one mechanism that produces apparent subadditivity in a one-component analysis, regardless of whether the underlying biology has a true direct BR-PB interaction.

The two mechanisms, namely true subadditive interaction and latent-class structure with class-correlated components, produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of PB that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components, $\hat{\gamma}$ is identified up to confounding with the class structure, and the analyst should report $\hat{\gamma}$ alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

3.3 Time-variant natural-history component (TV)

TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

Like PB, TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

3.3.1 When does natural history actually exist?

A common simplifying assumption in trial analysis is that subjects will improve over time, with or without treatment. This is true for some conditions and problematic for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

Acute conditions tend toward TV-positive: a sprained ankle, a post-operative pain syndrome, an episode of acute back pain all resolve on a timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial TV-improvement in the placebo arm. Chronic stable conditions (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward TV-zero on the population mean but with substantial individual variance. Chronic progressive conditions (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward TV-negative: a trial of a neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the slowing of progression, not absolute improvement. Cyclical and triggered conditions (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured TV that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

Selection-induced TV is real and rarely zero, even when the underlying condition is stable. Patients enrol at moments of clinical engagement that often coincide with recent worsening, so regression to the mean alone produces several points of apparent TV-improvement in the early weeks of a trial. As the introduction documents, the systematic-review evidence of[11],[12], and[13] (the last covering 202 trials across 60 conditions) and the antidepressant analyses of[6] and[7] show that much of what is conventionally attributed to placebo response is indistinguishable from regression to the mean and natural-history drift.

Trial participation itself also induces TV: the Hawthorne effect, the structured-visit effect, and the diary-completion effect appear in the data as TV-improvement that is independent of any pharmacological or placebo response.

3.3.2 Can we assume no TV effect?

But how safe is it to assume TV is zero? The short answer is: rarely safe, and never safe without empirical support. The longer answer is that TV may be assumed zero only if the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favor participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than assume it is zero.

The decomposition makes this assumption testable. A model that includes a TV component will return $\hat{m}_{TV}$ with a variance estimate; if both are small relative to $\hat{m}_{BR}$ and $\hat{m}_{PB}$, the analyst has empirical support for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others. Setting TV to zero and absorbing it into ε is worse than estimating it: the participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so absorbing it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

3.3.3 Why a Gompertz, not a linear time covariate?

A natural objection is that a model with `week` as a covariate already controls for time. But how adequate is such a control? Three reasons suggest it is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, BR and TV both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts discussed in section 4 are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

4 Identifiability and trial-design contrasts

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterize the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component, but only conditional on a known blinded-phase expectancy η: from a two-phase design just the product $\eta \cdot m_{PB}$ is identified, and separating η from m_{PB} requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields BR once drug carryover has elapsed, since PB and TV match across allocation arms within the blinded phase. All phases off drug yield TV plus any residual off-drug PB, with no active BR contribution beyond carryover and PB at the off-drug expectation level. The $\mathbf{1}_{\text{on drug}}$ indicator in the phase table above is shorthand for the continuous-decay drug state D_{it} defined in Table 1.

This conditioning on η is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design the data identify only the product $\eta \cdot m_{PB}$, so PB is recovered up to the assumed blinded-phase expectancy; the working value $\eta \approx 0.5$ is a modeling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for m_{PB} is accordingly the interval $\{\widehat{\eta m_{PB}}/\eta : \eta \in (0,1]\}$, which collapses to a point only when η is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating η from m_{PB}, and direct measurement of belief, as in the open-hidden and balanced-placebo designs[51,52] that manipulate or record the patient's treatment expectation. In a balanced-placebo design what the patient is told and what the patient receives are crossed, so that drug exposure and treatment belief vary independently. We therefore recommend that any trial relying on the decomposition for a PB-versus-BR attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to η across its plausible range.

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for BR, the open-label-versus-blinded contrast for PB, and an off-drug window for TV), and that off-drug window must be long enough relative to the carryover half-life for TV to separate from a still-decaying BR. Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at $N = 35$. We therefore do not assert that the three components are always recoverable. The pre-registered identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the BR estimate to the assumed η(phase) multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

5 Analysis-side methodology

In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

5.1 Four reasons not to match the DGP

The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation program) is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\widehat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz BR, PB, and TV simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant N, the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework[53], because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-treatment interaction, the $bm{:}D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarized through the continuous indicator D_{bc}, its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modeled exponential decay for residual carryover. The pharmacological-component target β_{bm}^{BR} is a more ambitious estimand: it is the biomarker-by-BR interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasized for individual treatment effects[49,54]: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

5.2 A small extension that usually pays off: phase indicators

The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
 Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase
+ (1 | ptID), correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR.

This extension adds three to four parameters at most, well within budget at any realistic N. It does not separately estimate BR and PB as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form. Where both the one-component and the phase-augmented analyses are reported, the additional `Dbc:phase` and `bm:Dbc:phase` terms introduce a small family of belief-attenuation tests, and the attribution inference they support should be treated as a pre-specified secondary analysis with its multiplicity acknowledged rather than as a second bite at the primary interaction test.

5.3 When to fit the full decomposition

A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterize how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

A word on the carryover parameterization is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator D_{bc} decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.[31] model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza[55] formalizes an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{”a}rtner et al.[56] benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The pre-registered simulation program of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

6 Worked example: information loss without decomposition

6.1 What the one-component analysis estimates

Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as $R_{it} = \mathrm{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$. The one-component analysis fits a single drug indicator D_{it}. In symbols:

$$R_{it} \;=\; \alpha + \beta_D \, D_{it} + g(t) + u_i + e_{it},$$

and reports $\hat{\beta}_D$ as 'the treatment effect'. Taking the on-drug minus off-drug contrast that β_D targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is:

$$\operatorname{plim} \hat{\beta}_D \;=\; m_{BR} + b_{PB} + b_{TV},$$

where $b_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$ is the belief differential between the contrasted phases and b_{TV} collects any natural-history contribution that fails to cancel under the contrast actually used; b_{TV} is zero for a clean on/off within-participant contrast and strictly positive for a baseline-anchored pre-post analysis, in which case $b_{PB} = E[PB]$ and $b_{TV} = E[TV]$. To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so $b_{TV} \approx 0$ there and the operative displacement is the belief term b_{PB}; the baseline-anchored pre-post form, which additionally carries $b_{TV} = E[TV]$, is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the one-component estimate over-states the pharmacological component m_{BR}. The displacement is a property of the probability limit and is unchanged as $N \to \infty$: a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker B_i. If that summary is the one-component per-participant effect, which estimates $m_{BR,i} + w_{PB}\, m_{PB,i} + w_{TV}\, m_{TV,i}$ for design-determined loadings $w_{PB}, w_{TV} \geq 0$, then by the linearity of covariance the fitted biomarker slope has probability limit:

$$\beta_{bm}^{\text{lumped}} \;=\; \frac{\operatorname{Cov}(m_{BR} + w_{PB} m_{PB} + w_{TV} m_{TV},\, B)}{\operatorname{Var}(B)} \;=\; \beta_{bm}^{BR} + w_{PB}\, \beta_{bm}^{PB} + w_{TV}\, \beta_{bm}^{TV},$$

where $\beta_{bm}^{C} = \operatorname{Cov}(m_C, B) / \operatorname{Var}(B)$ is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that β_{bm}^{PB} or β_{bm}^{TV} is non-zero, therefore registers a non-zero one-component slope even when its pharmacological association β_{bm}^{BR} is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the one-component slope from a genuine pharmacological one, because the two coincide in the estimand whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$.

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3, where the joint effect of BR and PB falls below their sum, a first-order correction proportional to $\mathrm{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the biomarker slope, and the identities above should be read as the leading behavior in the additive case rather than as exact under interaction. The empirical magnitude of that correction term is itself an open question that the framework's γ-interaction simulation (section 3.3) is designed to address.

Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand, $\operatorname{plim} \hat{\beta}_D$, not its sampling variability: the one-component coefficient remains a well-estimated number with a valid Wald test of its own nullity, and that test rejects at the rate dictated by the displaced value. A significance test of the one-component coefficient therefore cannot, in principle, detect the bias, because the bias does not move the coefficient toward zero. Detecting it requires comparing $\hat{\beta}_D$ against m_{BR}, which is exactly the comparison the decomposition, or a phase contrast that isolates m_{BR}, makes available and the one-component analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from m_{BR}. This is why the pilot of section 7.1 found the biomarker-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of $\hat{\beta}_{bm:D_{bc}}$, and a production run that means to test it must read the bias off the coefficient, not off the test.

6.2 A numerical illustration

To illustrate the inferential cost of failing to decompose, we shall work through a simple numerical example. Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values $m_{BR}^A = 4$, $m_{PB}^A = 6$, $m_{TV}^A = 1$. Participant B is a low placebo responder and a strong pharmacological responder, with $m_{BR}^B = 8$, $m_{PB}^B = 1$, $m_{TV}^B = 1$.

In the open-label phase (weeks 1 to 8), both BR and PB run at full strength and TV accumulates. By week 8 both participants have reached close to their saturation values:

| BR | $PB(\eta = 1)$ | TV | Total improvement | |—|—|—|—|—| |

Participant A | 4.0 | 6.0 | 1.0 | **11.0** | | Participant B | 8.0 | 1.0 | 1.0 | **10.0** |

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

In the blinded discontinuation phase (weeks 9 to 12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. For clarity of exposition we take the carryover half-life to be short relative to the four-week blinded window, so that BR has effectively cleared by the end of week 12; the more realistic case of a slowly decaying BR, modeled by the carryover machinery of section 3.1, attenuates but does not eliminate the contrast below. By the end of week 12, the components are:

| BR | $PB(\eta = 0.5)$ | TV | Total improvement | |—|—|—|—|—|
Participant A | 0.0 | 3.0 | 1.0 | **4.0** | | Participant B | 0.0 | 0.5 |

1.0 | **1.5** |

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits. For simplicity the table holds TV at its week-8 value; in reality TV continues to accumulate over weeks 9 to 12, which shifts both totals by a common amount and leaves the BR-versus-PB contrast on which the example turns unchanged.

A one-component fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B). - Blinded-placebo means: 4.0 (A), 1.5 (B). - Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75. - Estimated placebo effect: not identifiable from this contrast. - Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B; it estimates 'drug' at 7.75 points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no principled way to decompose further.

A linear-mixed-effects analysis with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. After fitting, the model returns $\hat{m}^A_{BR} = 4.0$, $\hat{m}^B_{BR} = 8.0$, $\hat{m}^A_{PB} = 6.0$, $\hat{m}^B_{PB} = 1.0$, $\hat{m}^A_{TV} = \hat{m}^B_{TV} = 1.0$ (at the population point estimate; participant-level estimates have meaningful uncertainty at N in the trial-relevant range, which the simulation study below characterizes).

The information loss from combining the components into one indicator is not abstract. We may compare the inferences directly:

Question | One-component answer | Three-component answer | |—|—|—| |

Does the drug work for A? | ~7.75 points (one-component). | $BR_A = 4.0$, distinct from her larger placebo response. | | Does the drug work for B? | ~7.75 points (one-component). | $BR_B = 8.0$, twice the magnitude of A. | | Are A and B different responders? | No discernible difference. | B has 2x A's BR but 1/6 of A's PB. | | Is a biomarker predicting BR useful here? | Cannot answer. | Yes if the biomarker correlates with $\hat{m}_{BR}$. | | Should A be continued long-term? | Cannot say. | Probably yes ($BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay. | | Population mean drug effect? | 7.75 points (biased). | $6.0 = (4.0 + 8.0)/2$ (correct mean of BR). |

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. Predictive-biomarker validation, which is a central scientific aim of the trial design family considered here, becomes impossible without the decomposition.

A second example at the population level illustrates the same problem. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug. Cohort 1 (early trial, naive patients) has mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$. Cohort 2 (later trial, patients who have had multiple prior failed treatments and are skeptical) has mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$. By construction the drug works equally well in both cohorts. But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug works less well in cohort 2 and might trigger a search for confounders or sub-population effects. None of this is real; the difference is entirely in PB, specifically in the population baseline level of placebo responsiveness. The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. Unfortunately, this is the kind of apparent non-replicability that pharmacology has spent the last decade struggling to navigate; the decomposition is one of the formal tools that makes it tractable.

7 Simulation study

The simulation study reported and pre-registered in this section is structured under the[57] ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The pre-registered plan is at `analysis/scripts/component-decomposition/00-ademp` A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

7.1 Pilot validation of analysis machinery (100 replicates, 6 cells)

The production simulation program is the empirical core of this manuscript; its first realized run, Study A, is reported in section 7.3, while the fuller pre-registered grid (section 7.2) remains partly outstanding. The pilot reported here precedes both and confirms only that the analysis machinery runs end-to-end.

A 100-replicate-per-cell pilot was run at the reference Hybrid open-label-plus-blinded-discontinuation (OL+BDC) design, $N = 35$, on the 6-cell slice $pb\_strength \in \{0, 6.5, 13\} \times$ analysis $\in$ {one-component, phase-augmented}, using 8-core `parallel::mclapply`. The convergence rate was 1.0 across all six cells. Per-replicate output is checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds` and the Morris-format cell-level summary at `analysis/data/quick-sim/06-decomp-summary.txt`. The exploratory summaries are diagnostic only. The one-component `bm:Dbc` test showed power 0.95, 0.93, and 0.83 across $pb\_strength \in \{0, 6.5, 13\}$, while the phase-augmented analysis showed 0.65, 0.65, and 0.61; the mean estimated $\widehat{\beta}_{bm:D_{bc}}$ was essentially flat within each analysis (about -2.25 for the one-component and -2.21 for the phase-augmented fit). At 100 replicates the MCSE on a power estimate near 0.7 is about 0.046, so these figures cannot by themselves separate the two analyses; their interpretation, including why the phase-augmented analysis loses its bias-detection contrast at this N, is resolved by the production run of section 7.3. No confirmatory bias or power claim is drawn from the pilot, and all substantive claims are deferred to the production program below.

Two design-level observations from the pilot inform the production specification. First, at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the D_{bc}:phase interaction term, removing the precise contrast that would have isolated the biomarker-treatment-by-phase signal the framework is designed to detect. Second, the predicted bias of the one-component analysis arises in the regression coefficient $\widehat{\beta}_{bm:D_{bc}}$, not in the indicator-level rejection rate, so power on the biomarker-treatment Wald test is not the right summary statistic to detect the framework's predicted bias. The production program below addresses both observations through larger N, a richer trial design that admits the full phase-augmented formula, and reporting of the coefficient mean against the implied true value rather than the rejection rate alone.

7.2 Production-simulation specification

The production program follows the[57] ADEMP framework and targets four deliverables: the bias of $\widehat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses across $N \in \{35, 70, 200, 300\}$ and a $pb\_strength$ grid; identifiability metrics for the full-decomposition fit (convergence, variance-inflation factors, and sensitivity of the BR estimate to the η(phase) multipliers); Type I error and power for the interaction test under an η-mismatch, TV-magnitude, and dropout sensitivity grid, with null cells at $n_{\text{reps}} =$ 5,000 for MCSE 0.003; and the participant-level interaction signal recovered as a function of (m_{BR}, m_{PB}, m_{TV}), which the one-component-total regression does not preserve. The `tidyverse` implementation collection drives the grid; driver scripts, checkpointed cell-level summaries, per-replicate output, and the committed ADEMP pre-registration all reside under `analysis/scripts/component-decomposition/`. The realized Study A run reported next used 1,000 replicates per alternative cell.

Two features of the sample-size axis should be read together with the results. First, N is the total across randomization paths, following the compendium convention, but the driver fixes the count passed to each path: the two smaller cells use a single-path design, so their totals are 35 and 70 as the driver states them, while the two larger cells use a two-path design and therefore analyze 200 and 300 participants where the driver grid records 100 and 150. The totals are quoted here. Second, that same threshold switches the design itself, from the eight-visit single-path form to the sixteen-visit two-path form with the crossover phase. The axis is consequently not a clean sample-size sweep, and the step between the 70 and 200 cells combines a larger sample, more visits per participant, and an added crossover contrast. Comparisons within either pair are interpretable; a comparison across the threshold is not attributable to sample size alone.

7.3 Study A results: bias and power at 1,000 replicates per alternative cell

The production Study A run (1,000 replicates per alternative cell, 5,000 per null cell, 80 alternative cells crossing $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 200, 300\}$, plus four null cells) returns a result that does not support the framework's central bias prediction. The DGP couples the baseline biomarker to the BR component only, through the on-drug biomarker-BR covariance channel, so the true biomarker-treatment effect is the biomarker-by-BR slope, $\beta_{bm:D_{bc}} = -c_{bm}\,\sigma_{BR}/\sigma_{bm} = -2.25$, and the biomarker is by construction uncorrelated with PB and TV. Under the one-component analysis the mean estimated $\hat{\beta}_{bm:D_{bc}}$ tracks this value across the whole grid: the mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 200$, and 0.006 at $N = 300$, in every case within one MCSE (respectively about 0.028, 0.019, 0.009, and 0.007) of zero, with no monotone trend across m_{PB} or m_{TV}. The predicted placebo-attribution bias of the biomarker-treatment estimate does not appear.

This is not a failure of the simulation; it is the behavior the identity of section 6.1 requires. That identity writes the one-component biomarker slope as $\beta^{BR}_{bm} + w_{PB}\beta^{PB}_{bm} + w_{TV}\beta^{TV}_{bm}$. With the biomarker coupled to BR alone, $\beta^{PB}_{bm} = \beta^{TV}_{bm} = 0$, so the one-component slope equals the pharmacological slope no matter how large m_{PB} or m_{TV} grows. Study A varies the magnitudes of the unmodeled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with PB or TV, which this DGP holds at zero. The clean null is therefore a confirmation of the identity and a correction to the looser claim of section 1 that large PB or TV by itself biases biomarker validation. It does not; biomarker-component coupling does.

The comparison between analyses is equally clear. The phase-augmented analysis is not less biased than the one-component analysis at any sample size (its mean absolute bias is 0.028 versus 0.016 at $N = 35$ and is comparable elsewhere), and it is markedly less powerful for the biomarker-treatment test: mean power across the grid is 0.35 versus 0.59 at $N = 35$ and 0.62 versus 0.90 at $N = 70$, with both analyses reaching essentially full power by $N = 200$. Because the DGP contains no biomarker-PB contamination for the phase contrast to remove, the extra phase-by-treatment terms add estimation cost without buying any bias reduction, and the analysis is dominated. Convergence was approximately complete for both analyses at every sample size (minimum convergence rate 0.998), so the rank-deficiency that forced dropping the `Dbc:phase` term in the $N = 35$ pilot, which we had advanced as the explanation for the pilot's inverted power ranking, is not operative in the production run: the inverted ranking persists with the full formula fitted and converged. The pilot's power inversion was therefore a genuine feature of this DGP, not an artifact of rank-deficiency.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of $\beta_{bm:D_{bc}}$ ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.98 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at $N = 200$ (0.056, MCSE 0.003) sits 1.9 standard errors above nominal but is within the simulation's Monte Carlo uncertainty.

Two conclusions follow for the Study A program. First, the result sharpens rather than supports the section 1 framing: the biomarker-validation failure is real as an identity, but it is driven by the biomarker's coupling to PB or TV, not by the magnitude of those components, and Study A as pre-registered varies the latter while holding the former at zero. The decisive axis is the biomarker-PB correlation $c_{bm,PB}$, which the contaminated-biomarker pilot in the following subsection varies directly. Second, on the present evidence the phase-augmented analysis carries a real power cost without a compensating bias reduction under an uncontaminated biomarker, so it should not be recommended as a default for this design; the pilot below evaluates whether contamination changes this verdict.

7.4 Contaminated-biomarker pilot (100 replicates per cell)

A supplementary pilot varying the biomarker-PB correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ at two PB magnitude levels ($m_{PB} \in \{0, 6\}$) and two sample sizes ($N \in \{70, 300\}$), using 100 replicates per cell, confirms the identity-based prediction and reveals an unexpected structural result.

At $N = 300$ the one-component bias scales monotonically with $c_{bm,PB}$: mean $\hat{\beta}_{bm:D_{bc}}$ moves from -2.31 ($c_{bm,PB} = 0$, bias -0.06) to -2.37 ($c_{bm,PB} = 0.15$, bias -0.12) to -2.08 ($c_{bm,PB} = 0.30$, bias $+0.17$) to -1.74 ($c_{bm,PB} = 0.45$, bias $+0.51$). The last value sits 30 MCSEs from the true -2.25, establishing the effect unambiguously. The direction of bias is positive (toward zero): high-$c_{bm,PB}$ participants experience stronger PB, which in the open-label phase adds to the apparent drug effect, but this elevation is partially absorbed by the bm main effect, leaving the $bm : D_{bc}$ interaction coefficient attenuated relative to the pure pharmacological value.

The unexpected result is that m_{PB} does not modulate the bias: at the contaminated cells ($c_{bm,PB} \geq 0.30$, $N = 300$) the $m_{PB} = 0$ and $m_{PB} = 6$ biases are identical to within 0.006, well inside Monte Carlo error, even as the bias itself grows to $+0.51$; the only larger discrepancy is at $c_{bm,PB} = 0$, where both biases are near zero and differ by 0.067 at the pilot's 100-replicate resolution. The driver is therefore $c_{bm,PB} \times \sigma_{PB}$, not $c_{bm,PB} \times m_{PB}$. Even when the population-mean PB amplitude is zero, individual participants have nonzero PB draws (standard deviation $\sigma_{PB} = 2$), and coupling the biomarker to that variance is sufficient to contaminate the estimator. A true no-PB control requires $\sigma_{PB} = 0$, not $m_{PB} = 0$. This refines the Study A design: varying m_{PB} at $c_{bm,PB} = 0$ correctly produced zero bias, but because the relevant null was $c_{bm,PB} = 0$, not $m_{PB} = 0$.

The phase-augmented analysis provides no protection: its bias at every $(c_{bm,PB}, m_{PB}, N)$ cell is indistinguishable from the one-component bias. Adding phase-by-treatment interactions to the model does not decompose the contamination because both analyses see the biomarker-PB covariance through the same open-label versus blinded-discontinuation contrast; the contrast changes the weight on PB (via expectancy) but does not isolate the PB component's contribution to the biomarker slope. Separating that contribution requires both a design that decouples drug exposure from belief and a decomposition that exploits the decoupling; Study B (section 7.5) shows that the standard hybrid design cannot achieve this separation, whereas a balanced-placebo design can.

The $N = 70$ results are uninformative at 100 replicates (Monte Carlo standard error ≈ 0.06, comparable to or larger than the bias at moderate $c_{bm,PB}$). The production run will use 1,000 replicates per cell to establish the $N = 70$ pattern and 5,000 null replicates to assess Type I error under contamination. Until the production run is complete, the quantitative bias estimates from this pilot should be treated as directionally informative but imprecise.

The pilot results reframe the paper's practical guidance. The risk of biomarker-validation failure in an aggregated N-of-1 study arises not from the presence of a strong placebo response in the population but from the biomarker's correlation with individual-level placebo responsiveness. In a population where the baseline biomarker is uncorrelated with each participant's placebo-belief response amplitude, the one-component estimator recovers the pharmacological interaction without bias, and the full decomposition machinery is not required. In a population where the biomarker covaries with psychological characteristics that predict placebo response, such as optimism, prior treatment experience, suggestibility, or expectancy at enrollment, the one-component analysis conflates pharmacological and placebo-belief prediction, and the scale of that conflation grows with the coupling strength. At $c_{bm,PB} = 0.45$, a value that equals the pharmacological coupling used throughout this paper, the one-component bias is approximately 0.51 points on the interaction scale, representing 23% of the true pharmacological effect. This is a practically meaningful distortion for any precision-medicine stratification decision.

The production run of the contaminated-biomarker study, at 1,000 alternative-cell replicates and 5,000 null-cell replicates, is required to confirm these estimates, establish the $N = 70$ pattern, and characterize Type I error under contamination. Results are pending.

7.5 Study B: recovery under a belief-decoupling design

Study B asks whether the contamination demonstrated in section 7.4 can be removed by decomposition. The answer is conditional on the trial design. The mechanism is that, in the multivariate normal (MVN) DGP, the biomarker-BR coupling acts only on drug (and so loads on the drug-state contrast D_{bc}), whereas the biomarker-PB coupling scales with the expectancy η (the placebo component's amplitude is proportional to η, so its biomarker-coupled part loads on η). A decomposition that interacts the biomarker with both D_{bc} and η can therefore isolate β^{BR}_{bm} on $bm{:}D_{bc}$, but only if D_{bc} and η vary independently in the design.

In the standard hybrid open-label-plus-blinded-discontinuation design they do not: on-drug time and belief accumulate together (their analysis-stage correlation is about $+0.6$), and we verified that the belief-covariate decomposition, a blinded-stratum drug contrast, and a Gompertz-basis component decomposition all fail to recover β_{bm}^{BR} on that design, performing no better than, and sometimes worse than, the one-component analysis. The remedy is a design that breaks the collinearity. A balanced-placebo (open-hidden) design[51,52] interposes a covert on-drug phase, in which the drug is administered without the participant's knowledge ($\eta = 0$), and an open-placebo phase, in which the participant is off drug but believes treatment continues ($\eta = 1$); the first supplies drug variation at fixed belief and the second belief variation at fixed drug exposure, reducing the D_{bc}-η correlation to about -0.66.

The production run (1{,}000 replicates per cell, balanced-placebo design with three open-label, six covert on-drug, and six open-placebo timepoints, $N \in \{70, 150\}$, $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$) confirms the recovery at full resolution. At $N = 150$ the mean estimated $\hat{\beta}_{bm:D_{bc}}$ from the decomposition tracks the pharmacological target of approximately -2.25 across the entire contamination range, with bias -0.022, $+0.008$, $+0.021$, and \`$rsprintf('$ at $c_{bm,PB} = 0, 0.1, 0.2, 0.3$ respectively, every value within one MCSE (about 0.02) of zero. The one-component estimator on the same data is unbiased only at $c_{bm,PB} = 0$ and develops a monotone bias of $+0.161$, $+0.340$, and $+0.476$ as the biomarker-PB correlation grows. Calibration tracks the bias: the decomposition holds near-nominal interval coverage (about 0.94 to 0.95), whereas one-component coverage degrades to 0.86 at $c_{bm,PB} = 0.3$. The $N = 70$ cells show the same pattern with the decomposition unbiased (absolute bias ≤ 0.08) and lower power (about 0.67 versus 0.93 at $N = 150$), reflecting the precision cost of the smaller sample and the extra belief-covariate terms. Convergence was complete in every cell (minimum observed convergence rate 1.000). Per-replicate output and the cell-level summary are at `analysis/data/derived_data/component-decomposition/study-b-balanced/`. The contamination range is bounded above by positive-definiteness: at $c_{bm,PB} = 0.45$ the biomarker would have to correlate 0.45 with both BR and PB, which the covariance cannot accommodate (minimum eigenvalue -0.35), so we restrict the production grid to the feasible range $c_{bm,PB} \leq 0.30$.

The result sharpens the paper's central claim in an important way. The inferential failure of section 7.4 is real and the decomposition does remove it, but the decomposition is not self-sufficient: on the standard hybrid design it does not recover the pharmacological slope, because the design does not supply independent variation in drug exposure and belief. The enabling condition is a design that decouples the two, of which the balanced-placebo and open-hidden paradigms[51,52] are the established instances. The practical recommendation that follows is specific: a biomarker-validation trial in which the candidate biomarker may correlate with placebo responsiveness should be fielded as a balanced-placebo or open-hidden design rather than the standard hybrid design, because only the former makes the pharmacological biomarker-treatment interaction separately identifiable. Three questions remain open. Recovery beyond the positive-definite contamination range is not characterized, since the MVN architecture cannot represent it; the participant-level component recovery that motivated the original identifiability aims of Study B is not addressed by the slope-level analysis here;

and whether the same recovery holds under the mean-moderation architecture, where the biomarker shifts component means rather than correlating with component draws, remains to be confirmed.

8 Application to predictive-biomarker validation

The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalized in the PATH statement[14]. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the one-component biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with PB and TV, and departs from it only by the biomarker's covariance with those components. The decomposition is therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

A direct regression of total response on baseline biomarker,

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the one-component total contains. If PB varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

The right approach is to regress each component on the biomarker:

$$
\begin{aligned}
BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\
PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\
TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}.
\end{aligned}
$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

The decomposition is therefore the mechanism by which the precision-medicine question is posed whenever the candidate biomarker may correlate with PB or TV. Where the biomarker can be assumed orthogonal to the non-pharmacological components, the identity of section 6.1 shows the one-component analysis already targets the biomarker-by-BR estimand, and the simulation of section 7.3 confirms it recovers that estimand without bias. The phase-augmented linear-mixed analysis introduced in section 5 retains a useful subset of the decomposition machinery at small N (the `bm:Dbc:phase` interaction tests whether the biomarker-treatment effect attenuates under blinding, the practical proxy for biomarker-by-PB contamination), but Study A shows this proxy earns its degrees of freedom only when such contamination is present: under an uncontaminated biomarker it reduced no bias and lost power. The full decomposition recovers the component-specific regressions explicitly when N permits and when the scientific question warrants the cost.

9 Discussion

The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7.3, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components' loading onto the contrast used, and, for a biomarker slope, only by the biomarker's covariance with PB and TV. Where those terms are zero, the design-contrast one-component analysis is already unbiased, and the production simulation confirms it: with a biomarker coupled to BR alone, the one-component biomarker-treatment estimate is unbiased across the whole PB-TV grid and the phase-augmented analysis is dominated. The framework's contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by-BR association, not the one-component total).

Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the η(phase) multipliers used to parameterize the expectancy modulation of PB are themselves modeling assumptions, and a sensitivity analysis varying η across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that[54] and[49] raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

A final point concerns the scope of the decomposition, where it is essential to distinguish the existence of the components from their identifiability. The claim that an observed longitudinal treatment response is the sum of a pharmacological, a belief-driven, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and in that sense the decomposition is not specific to N-of-1 trials. Identifiability, by contrast, is design-specific, and the distinction between population-level and participant-level recovery is decisive for the biomarker application. A standard parallel-group placebo-controlled trial supplies a between-arm placebo contrast that identifies the population-mean belief response but not a participant-level PB, and it contains no within-subject contrast in which belief varies independently of pharmacology; the participant-level biomarker-by-BR slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomized-withdrawal designs[51,52], the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends[1–3] operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

10 Conclusions

In conclusion, three points are to be emphasized. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of PB and TV: the one-component pre-post treatment effect is displaced by the PB and TV contributions, whereas the biomarker-treatment slope is displaced only by the biomarker's covariance with those components. Study A bears this out: varying PB and TV magnitude with a biomarker coupled to BR alone left the one-component estimate unbiased across the entire grid.

Second, the operative requirement is the design contrast, not the parametric analysis model. The trial design must include the drug-on/off and belief contrasts that identify the components, and the one-component analysis already exploits the drug-on/off contrast through its D_{bc} term. Adding parametric component structure on top is not free: under an uncontaminated biomarker the phase-augmented analysis reduced no bias and lost substantial power, so it should not be adopted as a default. Its value, and that of the full component-matched fit, is confined to the estimands where the extra structure does work, namely the attribution of response to pharmacology versus belief and the analysis of biomarkers that may themselves correlate with PB or TV.

Third, for predictive-biomarker validation the decomposition earns its place precisely when the candidate biomarker may correlate with PB or TV. A biomarker that predicts the one-component total is a pharmacological predictor only when it is uncorrelated with the non-pharmacological components; where that cannot be assumed, the one-component slope mixes pharmacological and non-pharmacological prediction, and the regulatory and prescribing decisions that motivate the trial require the separation the decomposition supplies. The contaminated-biomarker study (section 7.4) demonstrates this failure directly, and Study B (section 7.5) establishes that the bias is recoverable: a belief-covariate decomposition returns the unbiased pharmacological slope across the feasible contamination range, but only on a balanced-placebo design that decouples drug exposure from belief, not on the standard hybrid design. The decisive practical lesson is therefore as much about design as about analysis.

11 Conflict of interest

The authors declare no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

14 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds`; Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`. Driver scripts are under `analysis/scripts/component-decomposition/`. The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/06-component-decomposition/`.

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